Hence, if the exponential number is in the form 1^x or $[(\text{multiple of } 10)+1]^x$, then the unit digit is always 1, where x is a positive integer.

Therefore, the unit digit of 181^{47} is 1.

Similarly, by observing the following patterns, we can conclude that the unit digit of number with base ending with 5 is 5 and number with base ending with 6 is 6.

$5^1 = 5$ $\qquad$ $6^1 = 6$

$5^2 = 5 \times 5 = 25$ $\qquad$ $6^2 = 6 \times 6 = 36$

$5^3 = 25 \times 5 = 125$ $\qquad$ $6^3 = 36 \times 6 = 216$

Therefore, the unit digit of $55^4 = (50+5)^4$ is 5 and the unit digit of $56^{20} = (50+6)^{20}$ is 6.

We conclude that, for the base number whose unit digits are 0,1,5 and 6 the unit digit of a number corresponding to any positive exponent remains unchanged.

Example 3.11 Find the unit digit of the following exponential numbers:

(i) 25^{23} (ii) 81^{100} (iii) 46^{31}

Solution

(i) 25^{23}

Unit digit of base 25 is 5 and power is 23

Thus, the unit digit of 25^{23} is 5.

(ii) 81^{100}

Unit digit of base 81 is 1 and power is 100

Thus, the unit digit of 81^{100} is 1.

(iii) 46^{31}

Unit digit of base 46 is 6 and power is 31

Thus, the unit digit of 46^{31} is 6.

Try these

Find the unit digit of the following exponential numbers:

(i) 106^{21} (ii) 25^8

(iii) 31^{18} (iv) 20^{10}

Look into the following example. Observe the pattern of unit digit when the base is 4.

$4^1 = 4$ (odd power) $\qquad$ $4^2 = 4 \times 4 = 16$ (even power)

$4^3 = 4 \times 4 \times 4 = 16 \times 4 = 64$ (odd power) $\qquad$ $4^4 = 64 \times 4 = 256$ (even power)

$4^5 = 256 \times 4 = 1024$ (odd power) $\qquad$ $4^6 = 1024 \times 4 = 4096$ (even power)

Note that for base ending with 4, the unit digit of the expanded form alternates between 4 and 6. Further we can notice when the power is odd its unit digit is 4 and when the power is even it is 6.

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